

Financial Markets

Intermediate Macroeconomics

Fall 2025, Sciences Po. Paris Campus du Havre

Lectures 7.2

Motivation

- Financial markets play an essential role in the economy
- Our focus: money and interest rates
- What is the effect of **interest rates** on the macroeconomy?

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-
1. Today: focus on the **central bank (Federal Reserve Bank)**
 2. Next lecture: focus on banks

What is money?

- Lay English: “That person has a lot of money”
 - “That person is wealthy”
- “Money” has more specific meaning
 - Roughly: The asset that people use as a means of payment (post prices in terms of money)

Traditional Functions of Money

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3. Store of Value

- People may hold dollars so as to be able to buy something tomorrow or later

Demand for Money

Money and Bonds

Suppose you only have a choice between two assets: money and bonds.

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2. **Bonds**

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2. Bonds

- Pay a positive interest rate i , but cannot be used for transactions
- In reality there are many types of financial instruments that pay positive interest.
- Here: we assume only one group interest-bearing assets called “*bonds*”

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1. **Overall level of transactions in the economy:** is very hard to measure, but it is likely to be proportional to the nominal income ($\$Y$)
2. **Interest rates:** an increase in the interest rate decreases the demand for money, as people put more of their wealth into bonds

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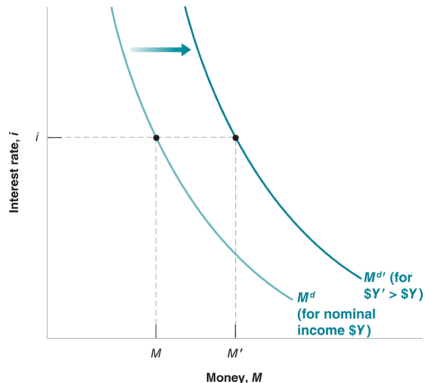
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Demand for money (M^d) is equal to nominal income $\$Y$ (a measure of level of transactions in the economy) times a decreasing function of the interest rate i

This equation means that the demand for money:

- Increases in proportion to nominal income
- Depends negatively on the interest rate.
 - An increase in the interest rate ($\nearrow i$) decreases the demand for money ($\searrow M^d$), as people put more of their wealth into bonds.

Demand for Money (M^d)



- For a given level of nominal income, a lower interest rate increases the demand for money.
- At a given interest rate, an increase in nominal income shifts the demand for money to the right.

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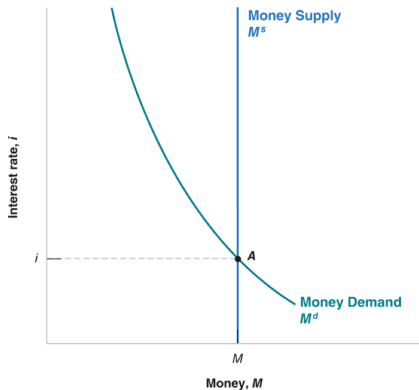
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- Money supply = Money demand

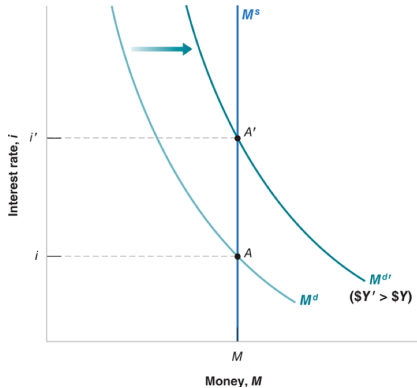
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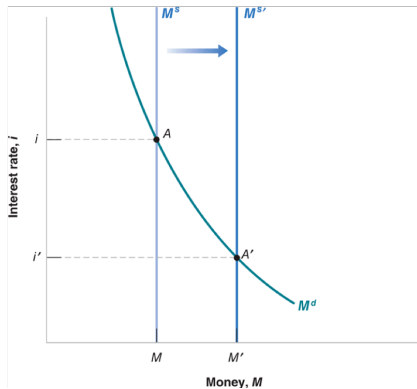
- The interest rate must be such that the supply of money (which is independent of the interest rate) is equal to the demand for money (which does depend on the interest rate).

The Effects of an Increase in Nominal Income on i



- Given the money supply, an increase in nominal income leads to an increase in the interest rate.

The Effects of an Increase in the Money Supply on i



- An increase in the supply of money leads to a decrease in the interest rate.

Summary

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- For a given M_s : an increase in nominal income ($\nearrow \$Y$) leads to an increase in the interest rate ($\nearrow i$)
- An increase in the supply of money by the central bank ($\nearrow M_s$) leads to a decrease in the interest rate ($\searrow i$)

Money supply in practice

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- **Expansionary OMOs:** the central bank expands the supply of money by buying bonds.
- **Contractionary OMOs:** the central bank contracts the supply of money by selling bonds.

The Balance Sheet of the Central Bank and OMOs

Balance Sheet	
Assets	Liabilities
Bonds	Money (currency)

The Effects of an Expansionary Open Market Operation	
Assets	Liabilities
Change in bond holdings: +\$1 million	Change in money stock: +\$1 million

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- An OMO in which the central bank buys bonds and issues money increases both assets and liabilities by the same amount.

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The price of the bond and the interest rate are negatively related

- The higher the interest rate, the lower the bond price today.

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- At a given interest rate, the Fed purchases bonds from the banks

Determining the Interest Rate: II

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- Step 1: we need to introduce banks

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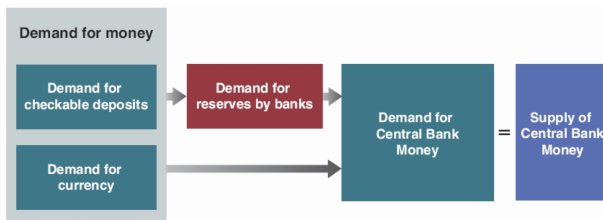
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 - The regulation states mandatory “minimum reserve requirements”
 - The central banks uses this to steer money supply in the economy

Balance Sheets: banks & the central bank

(a)	Central Bank	
	Assets	Liabilities
	Bonds	Central Bank Money = Reserves + Currency
(b)	Banks	
	Assets	Liabilities
	Reserves Loans Bonds	Checkable deposits

The demand and supply for central bank money

Intuition:



The demand for central bank money

1. Demand for currency (cash) by people:

2. Demand for reserves:

The demand for central bank money

1. **Demand for currency (cash) by people:** Today, we will simplify a lot
 - We will assume that people do not want to hold cash at all
 - People will only want to hold money as checkable deposits
 - Demand for currency (cash) by people = 0
 - Our results would not change if we allow for the demand for cash. Only the algebra becomes more complicated (See Appendix to Chapter 4)
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 - Demand for central bank money = demand for reserves by banks
 - Reserves depend on checkable deposits by people

The demand for central bank money

1. Demand for checkable deposits by people

- Our people don't hold cash
- Their demand for checkable deposits will be the same as last time for cash:

$$M^d = \$YL(i)$$

The demand for central bank money

2. What about banks' demand for reserves?

- Banks need to hold a fixed ratio θ of their deposits as reserves in the Fed
- θ = reserve ratio (e.g. if $\theta = 0.1$, banks need to hold 10% of their deposits as reserves)

$$H^d = \theta M^d$$

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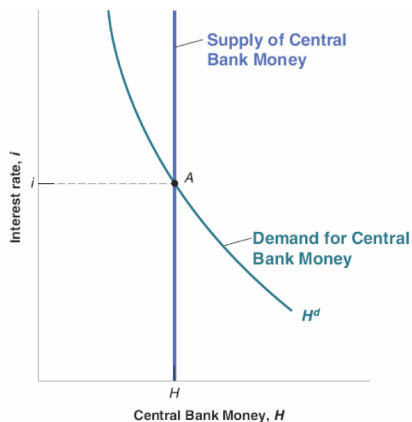
Equilibrium in the market for central bank money

- Let H denote the supply of central bank money
- Then the equilibrium condition:

$$H = H^d$$

$$H^d = \theta \$Y L(i)$$

Equilibrium in the market for central bank money



The equilibrium interest rate is such that the supply of central bank money is equal to the demand for central bank money.

Fed funds

- **The federal funds market** is an actual market for bank reserves
- **The federal funds rate**
 - is the interest rate determined in the federal funds market
 - is the main indicator of U.S. monetary policy because the Fed can choose the federal funds rate it wants by changing H .

Liquidity trap

The liquidity trap

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- When the interest rate is equal to zero, and people have enough money for transaction purposes, they become indifferent between holding money and holding bonds.

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More intuition:

- When the interest rate is equal to zero, and people have enough money for transaction purposes, they become indifferent between holding money and holding bonds.
- The demand for money becomes horizontal.
- This implies that, when the interest rate is equal to zero, further increases in the money supply have no effect on the interest rate, which remains equal to zero.

The liquidity trap

